

YENDLURI VIJAY VARDHAN

Data Science Engineer | GenAI & ML Specialist

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CAREER OBJECTIVE

Final-year B.Tech student in AI & Data Science seeking to build and deploy data-driven and GenAI solutions that improve business outcomes using robust ML practices, clear evaluation, and responsible AI guardrails, while continuously learning modern agentic workflows and MLOps on Azure.

TECHNICAL SKILLS

Programming	Python, SQL, Git
ML / Data	Pandas, NumPy, Scikit-learn, TensorFlow / Keras, OpenCV
NLP / LLMs	LangChain, Azure OpenAI, Prompt Engineering, RAG
Cloud / Tools	Azure AI Search, Azure Key Vault, Vector Search, MLOps (Azure)
Practices	Agentic Workflows, Tool/Function Calling, Evaluation Sets

INTERNSHIP EXPERIENCE

GenAI / Data Science Intern — Nayagara Technologies Limited

6 Months

Domain: Generative AI · E-Commerce

- ▶ Built a RAG-based GenAI assistant for product discovery using Azure OpenAI & LangChain.
- ▶ Implemented vector search, hybrid retrieval, and prompt engineering with citations.
- ▶ Improved retrieval accuracy via reranking and query rewriting techniques.
- ▶ Designed and delivered GenAI solutions using RAG patterns with grounded responses, citations, and fallback behavior.
- ▶ Implemented chunking, metadata filters, and evaluation sets to improve answer faithfulness and relevance.
- ▶ Built agentic workflows with tool/function calling, routing, and verification steps for multi-stage business tasks.
- ▶ Defined tool schemas and structured outputs to improve determinism and reduce hallucination risk.
- ▶ Engineered prompts for consistent JSON outputs and safe responses with content and policy guardrails.
- ▶ Applied prompt versioning and offline tests to measure quality, latency, and cost trade-offs.
- ▶ Developed classical ML classification models with robust validation, leakage checks, and threshold optimization.

PROJECTS

GenAI Shopping & Product Discovery Assistant (RAG)

Industry: Retail

Client: EComKart Retail (APAC) | Role: GenAI / Data Science Engineer

Tools: Python · Azure OpenAI · LangChain · Azure AI Search · Embeddings · Prompt Engineering · Key Vault

Built a GenAI assistant answering product questions and recommending alternatives using grounded retrieval over catalog, policy, and FAQ documents with citations and safety checks.

- ▶ Implemented RAG ingestion pipeline (catalog/FAQ), chunking strategies, and metadata-based filtering for relevance.
- ▶ Designed prompts for grounded answers with citations, refusal logic, and structured JSON outputs for UI rendering.
- ▶ Improved retrieval quality using hybrid search, reranking strategies, and query rewriting for ambiguous questions.
- ▶ Created offline eval sets (golden Q&A) and measured faithfulness/relevance to guide prompt and retrieval iterations.
- ▶ Integrated observability logs (latency/cost/error types) and established rollout checks for safe production use.

AI-Based Road Defect Classification System

Industry: Computer Vision / Transportation

Client: Smart Infrastructure Initiative (Academic) | Role: AI / ML Engineer

Tools: Python · TensorFlow/Keras · OpenCV · CNN · NumPy · Pandas · Matplotlib

Built an AI-powered system to detect and classify road defects (potholes, cracks, surface damage) from images to assist maintenance planning and smart city solutions.

- ▶ Developed and trained a CNN model to accurately classify road defects from images.
- ▶ Performed image preprocessing and augmentation to improve model performance and robustness.
- ▶ Built a data pipeline and evaluation system using accuracy, precision, recall, and F1-score metrics.
- ▶ Integrated the model into a frontend application for real-time image upload and defect classification.

EDUCATION

B.Tech in Artificial Intelligence & Data Science (Final Year)

2022 – 2026

Muthayammal Engineering College, Tamil Nadu

CERTIFICATIONS

- ▶ TCS NQT Certification – ION Digital Skills Platform
- ▶ NPTEL Certification – Foundation of Cloud IoT Edge ML
- ▶ NPTEL Certification – Cloud Computing (AWS)

DECLARATION

I hereby declare that all the information furnished above is true, complete, and correct to the best of my knowledge and belief.

Place: _____

Date: _____

(YENDLURI VIJAY VARDHAN)